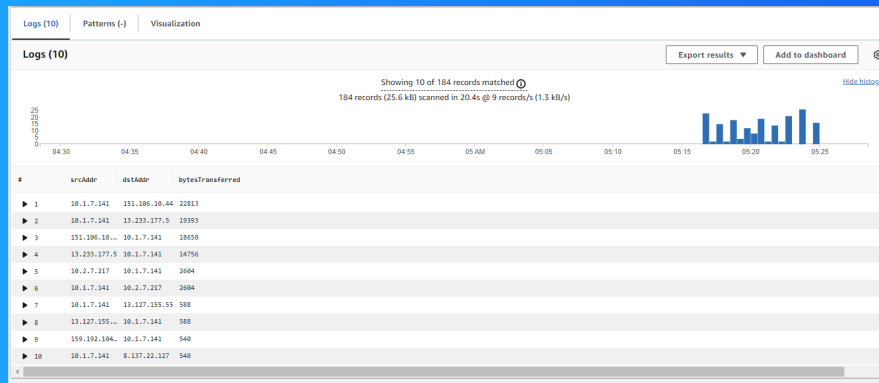




VPC Monitoring with Flow Logs



ajv.singh3107@gmail.com





Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a virtual private cloud service that allows you to create a virtual network in the AWS cloud. It is useful for isolating your resources and controlling network traffic within your VPC.

How I used Amazon VPC in this project

I used Amazon VPC to create isolated networks for our resources, allowing secure communication between instances. I configured subnets, route tables, and peering connections to enable seamless data flow while maintaining security and performance.

One thing I didn't expect in this project was...

I didn't expect in this project was the complexity of configuring route tables and ensuring proper traffic flow between VPCs. I underestimated the level of detail needed to troubleshoot connectivity issues and optimize network performance.

This project took me...

This project took me about 2 hours to complete, including setting up the VPCs, configuring the peering connection, updating route tables, and analyzing the flow logs. It required careful attention to detail to ensure everything functioned smoothly.



In the first part of my project...

Step 1 - Set up VPCs

In this step, I'll create two new VPCs to simulate separate networks. This is the foundation for setting up VPC peering, which will allow secure communication between these networks for testing and analysis later.

Step 2 - Launch EC2 instances

In this step, we'll launch EC2 instances in each VPC to test the VPC peering connection. These instances will send and receive data, allowing us to monitor network activity and ensure proper communication between resources in both VPCs.

Step 3 - Set up Logs

I'll set up VPC Flow Logs to capture all inbound and outbound network traffic. I'll also create a CloudWatch log group to store these records. This helps monitor network activity, detect issues, and analyze traffic patterns for better security.

Step 4 - Set IAM permissions for Logs

I'll create an IAM policy and role to give VPC Flow Logs the necessary permissions to write logs and send them to CloudWatch. This ensures that network traffic data is captured and stored for analysis, completing the flow log setup for my subnet.



Multi-VPC Architecture

I started my project by launching two VPCs. The first one, NextWork-1, has an IPv4 CIDR block of 10.1.0.0/16. The second, NextWork-2, has a unique IPv4 CIDR block of 10.2.0.0/16. I haven't created subnets yet, but they'll be added in the next step.

The CIDR blocks for VPCs 1 and 2 are 10.1.0.0/16 and 10.2.0.0/16. They have to be unique because overlapping IP ranges would prevent proper routing between the VPCs, causing conflicts in communication, especially when setting up peering connections.

I also launched EC2 instances in each subnet

My EC2 instances' security groups allow inbound SSH (port 22) and ICMP (ping) from the other VPC, and outbound traffic is unrestricted. This is because I need secure access between the instances for testing connectivity and monitoring network traffic



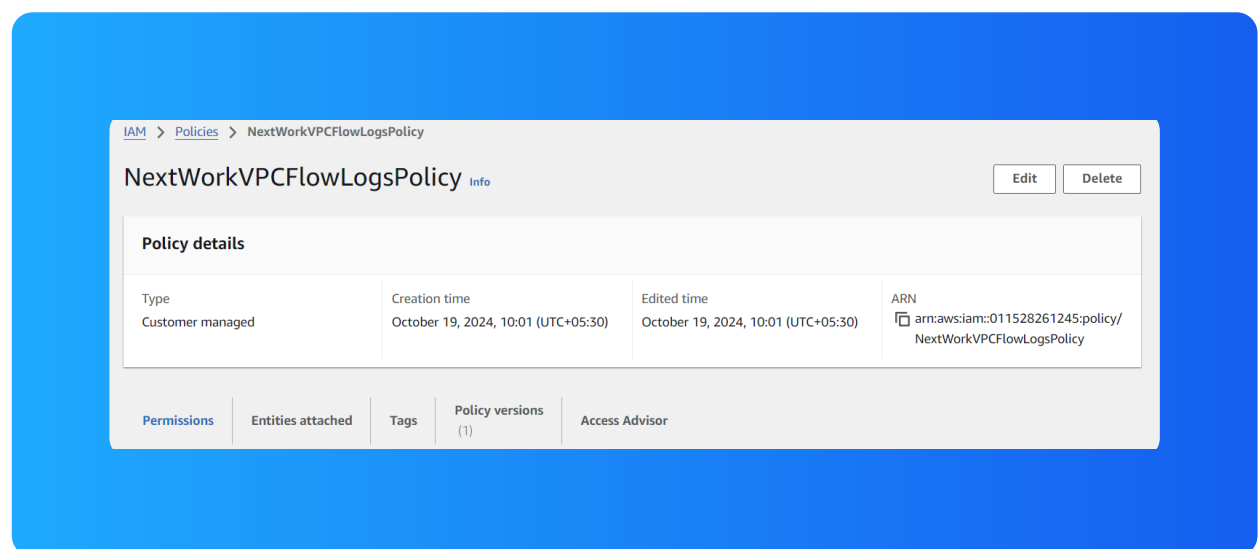


Logs

Logs are records of events or activities generated by AWS services, applications, or network traffic. They provide detailed information for monitoring, troubleshooting, and analyzing system performance and security in the cloud environment.

Log groups are containers in AWS CloudWatch that organize and store related log streams. They help manage logs from different resources, making it easier to view, search, and analyze logs for monitoring and troubleshooting purposes.

I also set up a flow log for VPC 1



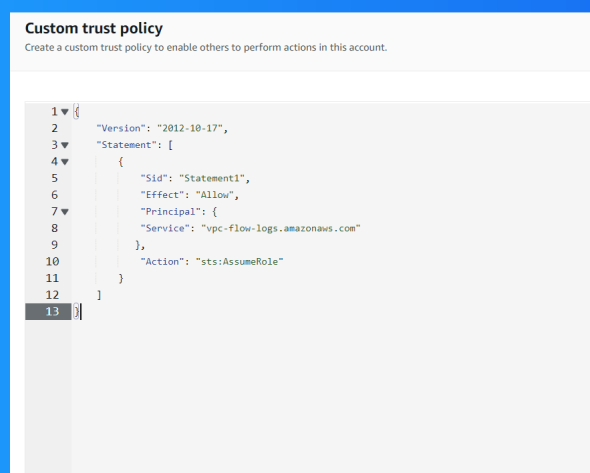


IAM Policy and Roles

I created an IAM policy because it specifies the permissions VPC Flow Logs need to write logs to CloudWatch. This ensures secure and effective operation, enabling me to monitor and analyze network traffic without compromising access control.

I also created an IAM role because it allows VPC Flow Logs to assume the necessary permissions defined in the IAM policy. This role securely grants the service access to write logs to CloudWatch, enabling effective monitoring of network traffic.

A custom trust policy is a set of permissions that specifies which entities (like AWS services or accounts) are allowed to assume an IAM role. It defines the relationship between the role and trusted entities, enabling secure access to resources.





In the second part of my project...

Step 5 - Ping testing and troubleshooting

I'll generate network traffic by having Instance 1 in VPC 1 send test messages to Instance 2 in VPC 2. This tests our VPC peering setup and verifies that the flow logs capture the traffic correctly, ensuring everything works as intended.

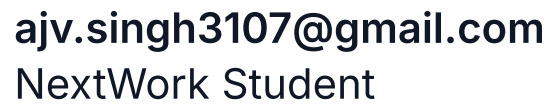
Step 6 - Set up a peering connection

In this step, we'll create a peering connection between our VPCs to enable communication. This will resolve connectivity issues by allowing resources in one VPC to access those in the other. Let's set up the peering connection and configure the route

Step 7 - Update VPC route tables

Step 8 - Analyze flow logs

In this step, we'll review VPC Flow Logs for VPC 1's public subnet to monitor network activity. Analyzing these logs will help us identify traffic patterns, troubleshoot issues, and optimize resource usage for a healthy network environment.

[illegible]

The ping test was successful, indicating that there is no network connectivity issue. The problem likely lies with the security group rules or the target instance's configuration. I will investigate further.



Connectivity troubleshooting

The ping test failed because there was no route defined for Instance 2's private IP address range in VPC 1's route table. I will create a route to resolve this.

To solve this, I set up a peering connection between my VPCs

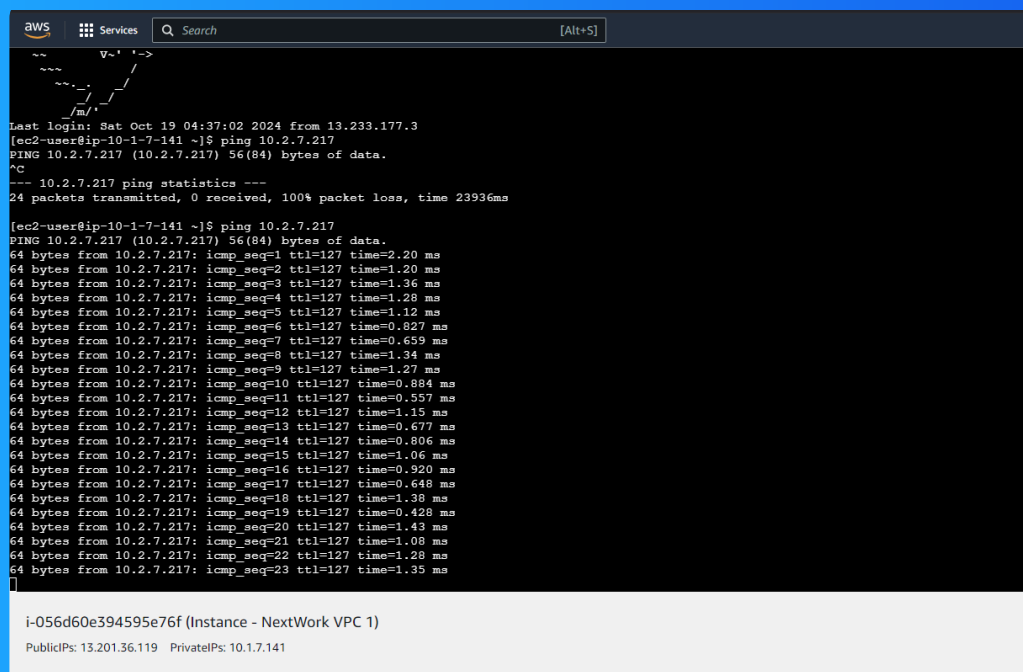
I also updated both VPCs' route tables so that traffic can flow between them via the new peering connection. This ensures that resources in each VPC can communicate properly, allowing for seamless data exchange and enhancing overall functionality.

rtb-01521362e630546cb / NextWork-1-rtb-public				Actions
Details				
Route table ID	rtb-01521362e630546cb	Main	No	Explicit subnet associations
VPC	vpc-0992d249b0401703c NextWork-1-vpc	Owner ID	011528261245	subnet-073f8884b6d6580e1 / NextWork-1-subnet-public-1-ap-south-1a
Routes				
Subnet associations Edge associations Route propagation Tags				
Routes (2)				
Filter routes				
Destination	Target	Status	Propagated	
10.1.0.0/16	local	Active	No	
10.2.0.0/16	pcx-08f428f0436f9a914	Blackhole	No	



Connectivity troubleshooting

I received ping replies from Instance 2's private IP address! This means the peering connection is functioning correctly, and the VPCs can communicate with each other. It confirms that our network setup is now allowing data exchange as intended.



```
aws
Services
Search
[Alt+S]

Last login: Sat Oct 19 04:37:02 2024 from 13.233.177.3
[ec2-user@ip-10-1-7-141 ~]$ ping 10.2.7.217
PING 10.2.7.217 (10.2.7.217) 56(84) bytes of data.
^C
--- 10.2.7.217 ping statistics ---
24 packets transmitted, 0 received, 100% packet loss, time 23936ms

[ec2-user@ip-10-1-7-141 ~]$ ping 10.2.7.217
PING 10.2.7.217 (10.2.7.217) 56(84) bytes of data.
64 bytes from 10.2.7.217: icmp_seq=1 ttl=127 time=2.20 ms
64 bytes from 10.2.7.217: icmp_seq=2 ttl=127 time=1.20 ms
64 bytes from 10.2.7.217: icmp_seq=3 ttl=127 time=1.36 ms
64 bytes from 10.2.7.217: icmp_seq=4 ttl=127 time=1.28 ms
64 bytes from 10.2.7.217: icmp_seq=5 ttl=127 time=1.12 ms
64 bytes from 10.2.7.217: icmp_seq=6 ttl=127 time=0.827 ms
64 bytes from 10.2.7.217: icmp_seq=7 ttl=127 time=0.659 ms
64 bytes from 10.2.7.217: icmp_seq=8 ttl=127 time=1.34 ms
64 bytes from 10.2.7.217: icmp_seq=9 ttl=127 time=1.27 ms
64 bytes from 10.2.7.217: icmp_seq=10 ttl=127 time=0.884 ms
64 bytes from 10.2.7.217: icmp_seq=11 ttl=127 time=0.557 ms
64 bytes from 10.2.7.217: icmp_seq=12 ttl=127 time=1.15 ms
64 bytes from 10.2.7.217: icmp_seq=13 ttl=127 time=0.677 ms
64 bytes from 10.2.7.217: icmp_seq=14 ttl=127 time=0.806 ms
64 bytes from 10.2.7.217: icmp_seq=15 ttl=127 time=1.06 ms
64 bytes from 10.2.7.217: icmp_seq=16 ttl=127 time=0.920 ms
64 bytes from 10.2.7.217: icmp_seq=17 ttl=127 time=0.648 ms
64 bytes from 10.2.7.217: icmp_seq=18 ttl=127 time=1.38 ms
64 bytes from 10.2.7.217: icmp_seq=19 ttl=127 time=0.428 ms
64 bytes from 10.2.7.217: icmp_seq=20 ttl=127 time=1.43 ms
64 bytes from 10.2.7.217: icmp_seq=21 ttl=127 time=1.08 ms
64 bytes from 10.2.7.217: icmp_seq=22 ttl=127 time=1.28 ms
64 bytes from 10.2.7.217: icmp_seq=23 ttl=127 time=1.35 ms
^C

```

i-056d60e394595e76f (Instance - NextWork VPC 1)

PublicIPs: 13.201.36.119 PrivateIPs: 10.1.7.141



Analyzing flow logs

Flow logs tell us about the source and destination IP addresses, traffic type, protocols used, ports, and the number of bytes transferred. They also include timestamps, allowing us to track when the traffic occurred for better analysis.

The flow log I've captured tells us there was successful traffic between specific IP addresses, indicating effective communication. It also shows the protocols used, port numbers, and data volume, helping us assess network performance.

The screenshot shows the AWS CloudWatch console interface for the log group 'NextWorkVPCFlowLogGroup'. The 'Log events' tab is active, displaying a list of log events. The table has two columns: 'Timestamp' and 'Message'. The messages are JSON-formatted flow log records. The first few records show 'REJECT' status, while later records show 'ACCEPT' status, indicating successful traffic.

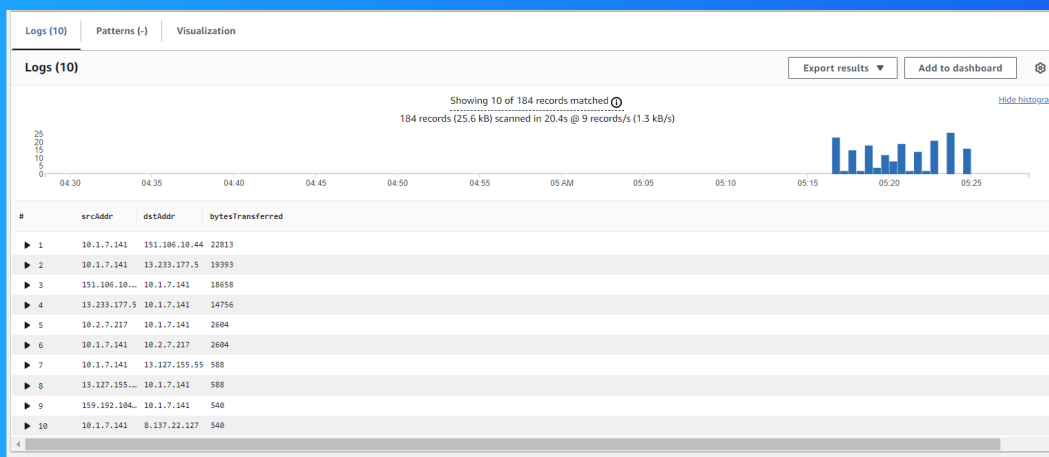
Timestamp	Message
2024-10-15T01:16:00Z	2 HELIXE2C2451 vpc-80777f59d0f4e204 51.6.229.104 10.1.7.141 54400 2885 6 1 40 172010866 172010861 REJECT DN
2024-10-15T01:16:00Z	2 HELIXE2C2451 vpc-80777f59d0f4e204 179.43.137.142 10.1.7.141 47129 31607 8 1 40 172010869 172010861 REJECT DN
2024-10-15T01:16:00Z	2 HELIXE2C2451 vpc-80777f59d0f4e204 181.224.128.83 10.1.7.141 34789 8000 6 1 40 172010868 172010861 REJECT DN
2024-10-15T01:16:00Z	2 HELIXE2C2451 vpc-80777f59d0f4e204 95.280.220.194 10.1.7.141 55278 9500 6 1 40 172010868 172010861 REJECT DN
2024-10-15T01:16:00Z	2 HELIXE2C2451 vpc-80777f59d0f4e204 185.224.100.185 10.1.7.141 48123 3995 6 1 40 172010869 172010861 REJECT DN
2024-10-15T01:16:00Z	2 HELIXE2C2451 vpc-80777f59d0f4e204 102.256.240.61 10.1.7.141 10046 8947 6 0 40 172010868 172010861 REJECT DN
2024-10-15T01:16:00Z	2 HELIXE2C2451 vpc-80777f59d0f4e204 146.185.125.76 10.1.7.141 40712 5212 6 1 40 172010868 172010861 REJECT DN
2024-10-15T01:16:00Z	2 HELIXE2C2451 vpc-80777f59d0f4e204 205.210.151.12 10.1.7.141 54191 444 6 1 40 172010869 172010861 REJECT DN
2024-10-15T01:16:00Z	2 HELIXE2C2451 vpc-80777f59d0f4e204 161.226.100.147 10.1.7.141 56668 9551 6 1 40 172010868 172010861 REJECT DN
2024-10-15T01:16:00Z	2 HELIXE2C2451 vpc-80777f59d0f4e204 89.248.101.142 10.1.7.141 44072 38114 6 1 40 172010868 172010861 REJECT DN
2024-10-15T01:16:00Z	2 HELIXE2C2451 vpc-80777f59d0f4e204 151.186.100.44 10.1.7.141 13386 22 6 3 100 172010869 172010861 ACCEPT DN
2024-10-15T01:16:00Z	2 HELIXE2C2451 vpc-80777f59d0f4e204 10.1.7.141 151.186.100.44 12 13384 6 6 400 172010868 172010861 ACCEPT DN
2024-10-15T01:16:00Z	2 HELIXE2C2451 vpc-80777f59d0f4e204 161.226.100.194 10.1.7.141 50018 235 6 1 40 172010868 172010861 REJECT DN
2024-10-15T01:16:00Z	2 HELIXE2C2451 vpc-80777f59d0f4e204 154.233.100.198 10.1.7.141 46219 3698 6 1 40 172010869 172010861 REJECT DN
2024-10-15T01:16:00Z	2 HELIXE2C2451 vpc-80777f59d0f4e204 51.101.101.100 10.1.7.141 28019 9151 6 1 40 172010868 172010861 REJECT DN
2024-10-15T01:16:00Z	2 HELIXE2C2451 vpc-80777f59d0f4e204 43.86.89.3 10.1.7.141 63941 504 6 1 12 172010868 172010861 REJECT DN



Logs Insights

Logs Insights is a powerful tool that enables us to query and analyze log data in real time. It helps us identify patterns, troubleshoot issues, and gain insights into system performance by allowing us to run custom queries and visualize log data.

I ran the query `stats sum(bytes) as bytesTransferred by srcAddr, dstAddr | sort bytesTransferred desc | limit 10``. This query analyzes VPC flow logs to identify the top 10 byte transfers by source and destination IP addresses.





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